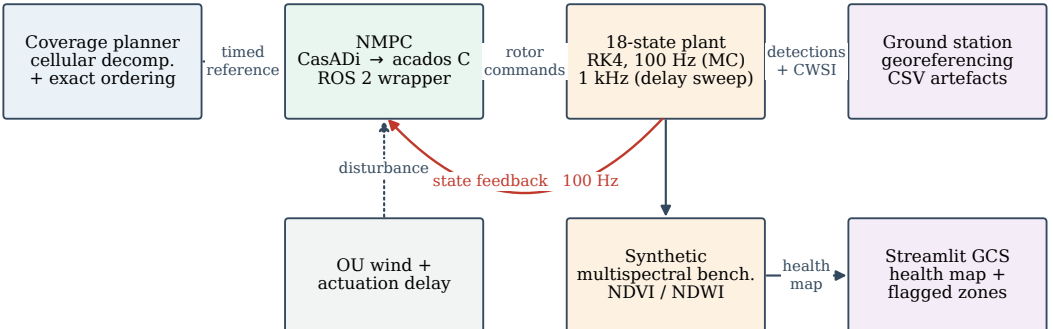
Coverage planner
cellular decomp.
+ exact ordering
timed
reference
NMPC
CasADi → acados C
ROS 2 wrapper
rotor
commands
18-state plant
RK4, 100 Hz (MC)
1 kHz (delay sweep)
detections
+ CWSI
Ground station
georeferencing
CSV artefacts
disturbance
state feedback
100 Hz
OU wind +
actuation delay
Synthetic
multispectral bench.
NDVI / NDWI
health
map
Streamlit GCS
health map +
flagged zones